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Experimental investigation of flow boiling heat transfer in an annular minichannel with ZnO-, ZnO/PMHS-, and Al_2O_3-modified heated surfaces

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Abstract: Subcooled flow boiling of distilled water was investigated in a vertical annular minichannel with interchangeable smooth and surface-modified heated tubes. Copper and stainless-steel substrates were tested before and after modification with ZnO or ZnO/PMHS; an Al_2O_3 coating was additionally tested on stainless steel. A simplified one-dimensional cylindrical model was used to correct thermocouple temperatures for radial conduction and determine the local effective heat transfer coefficient at the wall–fluid interface. Modified surfaces were compared pointwise with the corresponding smooth references at matched operating conditions and axial positions. A modification was considered beneficial only when it increased the heat transfer coefficient and simultaneously reduced, or at least did not increase, the wall temperature. ZnO on stainless steel was the only modification that consistently met this criterion. The mean pointwise heat transfer coefficient increase was 44.2% at 7 kg/h and 42.9% at 10 kg/h, with median increases of 37.8% and 36.8%, respectively. Mean wall temperature decreased by 36.9 K and 32.8 K. Al_2O_3 and ZnO/PMHS on stainless steel reduced the coefficient and increased wall temperature. The copper modifications showed no robust improvement, while their assessment was additionally affected by run-to-run variability in the smooth copper reference data. The results identify ZnO-modified stainless steel as the most effective tested surface.

Keywords: flow boiling; annular minichannel; heat transfer coefficient; ZnO coating; ZnO/PMHS; Al_2O_3 coating; distilled water

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1. Introduction

Flow boiling in mini- and microchannels is one of the most effective methods for removing high heat fluxes from compact thermal systems. The reduction of hydraulic diameter increases the surface-to-volume ratio and enables high heat-transfer intensity in a small volume. At the same time, the confined geometry makes the process sensitive to surface condition, channel shape, local vapour generation, pressure drop and liquid replenishment. In annular minichannels, wall curvature and the narrow annular gap additionally affect bubble growth, local liquid-film distribution and development of the thermal boundary layer.

Surface modification is a direct route to controlling boiling heat transfer. Modified surfaces can change the onset of nucleate boiling, the density of active nucleation sites, bubble departure, rewetting behaviour, critical heat flux and wall-temperature stability. The effect is not universal. A surface that promotes nucleation can also increase vapour retention, while a strongly wettable surface can improve liquid supply without necessarily providing sufficient nucleation activity. The thermal response therefore depends on the combined action of wettability, roughness, morphology, coating stability, substrate material and flow conditions.

Hydrophilic and superhydrophilic surfaces generally favour liquid spreading and rewetting of the heated wall, which may suppress local dry spots and stabilise heat transfer in confined channels. Hydrophobic and superhydrophobic surfaces can reduce the energy barrier for vapour embryo formation and promote earlier nucleation, but excessive hydrophobicity or vapour entrapment within surface texture can reduce the effective liquid-solid contact area and increase wall temperature. Hydrophobicity should therefore not be treated as intrinsically favourable in flow boiling.

The biphilic or mixed-wettability concept is particularly relevant because hydrophobic domains can act as preferential nucleation sites, while hydrophilic regions maintain liquid supply and restrict vapour spreading. This approach has been demonstrated mainly in pool boiling and selected microchannel studies. Its transfer to annular minichannels is nontrivial because the annular gap imposes different constraints on liquid and vapour transport.

The present study investigates subcooled flow boiling of distilled water in an annular minichannel. Smooth reference surfaces and three classes of modification were considered: ZnO, ZnO/PMHS and Al_2O_3. The analysis focuses on a corrected cylindrical data-reduction model and differential comparison with smooth references. The main objective was to determine whether the tested modified heated surfaces provide a measurable thermal advantage over the corresponding smooth substrate. The primary indicators were the local heat transfer coefficient, wall temperature and axial heat-transfer profiles obtained during increasing and decreasing heater power.

The results were analysed in a differential manner. For each substrate material and each mass flow rate, the smooth surface was treated as the reference case. Each modified surface was compared with the corresponding smooth reference at the same nominal heater-power setting, power branch, and axial position. This approach reduces the risk of overinterpreting absolute heat transfer coefficients that may be affected by baseline variability, operating conditions, substrate material, thermal contact and local wall-temperature distribution.

An effective modification was defined as one that increases the heat transfer coefficient and simultaneously reduces, or at least does not increase, the wall temperature. A surface producing a lower wall temperature without an increase in the coefficient was not treated as an clearly heat-transfer-enhancing surface. On the other hand, an isolated increase in the coefficient was not considered sufficient evidence of robust enhancement unless it was accompanied by a consistent wall-temperature response over the analysed range. The present distilled-water series establishes an initial ranking of coating-substrate combinations and defines priorities for future refrigerant experiments.

2. Literature background

A comprehensive review by Liang and Mudawar [1] shows that channel flow-boiling enhancement by surface modification can improve heat transfer coefficient and critical heat flux, but it can also increase pressure drop or introduce flow instabilities. This is important for minichannel design, where enhancement must be assessed using thermal, hydraulic and repeatability criteria rather than by the maximum coefficient only.

Betz et al. [2] demonstrated that the best pool-boiling performance is not necessarily obtained on uniformly superhydrophilic or uniformly superhydrophobic surfaces. Biphilic surfaces combining hydrophilic and hydrophobic regions can manage liquid and vapour transport more effectively. Although that work concerned pool boiling, the underlying requirement of simultaneous nucleation promotion and rewetting is directly relevant to confined flow boiling.

Phan et al. [3] investigated flow boiling of water on nanocoated surfaces in a single rectangular microchannel and showed that wettability strongly affects the heat transfer coefficient. Ahmadi et al. [4] examined flow boiling on mixed-wettability surfaces with gradient biphilicity in a high-aspect-ratio microchannel, supporting the view that wettability patterning can affect flow patterns and heat transfer. These studies provide useful reference points for future annular-minichannel work with deliberately patterned hydrophilic-hydrophobic surfaces.

Wciślik et al. [5] examined contact-angle determination using the sessile-drop method and an optical goniometer. They showed that the measured contact angle depends strongly on droplet dosing conditions and on the selected profile-fitting model, comparing circle, ellipse and Young–Laplace approaches. The Young–Laplace method was indicated as the most reliable option for non-ideal droplet shapes.

Kaniowski and Wciślik [6] analysed the influence of contact angle and surface tension on pool boiling on copper surfaces. Contact angle and surface tension were measured for deionised water, ethanol, FC-72 and Novec-649 using the sessile-drop method with a Krüss DSA25E goniometer and the Young–Laplace model. They also compared copper surfaces before and after heating, showing that surface condition and fluid properties strongly affect wettability and, consequently, boiling behaviour.

Surface modification can be achieved not only by depositing functional coatings but also by direct laser-beam structuring, or by combining both approaches. Radek et al. [7] examined laser processing of electro-spark-deposited WC-Co coatings on C45 steel and showed that it substantially changed the surface topography and residual-stress state while improving coating adhesion and corrosion resistance. Orman et al. [8] used laser treatment to generate microfins and roughened groove surfaces for pool-boiling enhancement. The strongest response was obtained for the highest microfins and roughest surfaces, particularly at low wall superheats; for the tested geometries, the heat flux from surfaces with 0.4 mm microfins was more than three times higher for water and twice as high for ethanol than that from surfaces with 0.2 mm microfins. Plasma-sprayed composite coatings provide another route to tailoring functional surface properties. Żórawski et al. [9] investigated Al_2O_3–$3TiO_2$ composite coatings containing CaF_2 and showed that plasma-sprayed coating composition can be used to reduce the friction coefficient and modify tribological performance. Taken together, these studies demonstrate that coating deposition and laser processing can control surface composition and morphology over multiple length scales, providing flexible ways for modifying interfacial and functional surface properties.

Studies directly concerning annular minichannels are also relevant to the present study. Single-phase studies also confirm that narrow annular geometry produces thermal behaviour different from that observed in conventional channels. Lu and Wang [10] experimentally investigated single-phase convective heat transfer of water in a narrow annulus for horizontal, upward and downward flow orientations. They found that the thermal behaviour of the narrow annulus differed from that of conventional circular tubes, particularly through an earlier transition from laminar to turbulent heat transfer. At Reynolds numbers below 150, axial heat conduction significantly affected the overall thermal response and contributed to heat-transfer deterioration. The study also demonstrated that the calculated heat-transfer characteristics depended on the data-processing method, with particularly large discrepancies between the wall-temperature and separation-coefficient approaches at Reynolds numbers below 400. More recently, Yang et al. [11] combined experiments and CFD for water flow in a horizontal mini-annular channel with a 3.0 mm gap. Their results showed turbulent behaviour within the Reynolds-number range of 1100–1729, relatively uniform cross-sectional temperature and velocity fields, and increasing Nusselt number with Reynolds number, heat flux and inlet temperature. These findings demonstrate that annular gap dimensions and operating conditions require geometry-specific thermal and hydraulic assessment.

The preparation and interpretation of the modified surfaces can also be supported by previous studies on coating-controlled vapour-layer behaviour [12,13]. Maziukienė et al. [12] investigated Leidenfrost-phenomenon formation on TiO_2-coated surfaces and modelled the associated heat-transfer processes. Vorotinskienė et al. [13] analysed the effect of ZnO tetrapod coating hydrophobicity on vapour-film formation and friction reduction. In the present study, Al_2O_3 was introduced as an additional ceramic-oxide surface variant on steel. ZnO nanowires are of significant importance because their tunable wettability, governed by their crystal structure and surface morphology, enables a wide range of multifunctional applications, including self-cleaning, oil–water separation, antifogging, antibacterial surfaces, and other advanced technologies [14].

Hożejowska and Piasecka [15] developed a mathematical model for flow boiling heat transfer in an annular minigap by formulating the problem in cylindrical coordinates, assuming axisymmetric steady-state conduction, and applying a third-kind boundary condition at the fluid–heater interface. In the present work, this mathematical formulation is simplified and adapted to the current test section, where the data reduction is restricted to a local radial-conduction correction in the coaxial heated metal element adjacent to the cartridge heater.

Experimental studies have shown that flow boiling in annular gaps depends on gap width, mass flux, heat flux and local subcooling [16,17]. Enhanced-wall studies in annular minigaps have shown that surface structure can alter local heat transfer and boiling development [18]. Recent annular-minigap experiments with HFE-649 further demonstrated that both flow resistance and heat transfer must be analysed jointly when interpreting boiling behaviour in this geometry [19].

3. Materials and methods

3.1. Experimental facility and annular minichannel geometry

The experimental setup consisted of a closed-loop circulation system, an annular-minichannel test module, a heating and power-supply system, and a data-acquisition system. A schematic diagram of the experimental rig is shown in Fig. 1a, whereas Fig. 1b presents a photograph of part of the laboratory stand, including the test-section assembly. Distilled water was used as the working fluid. The flow was generated by a Tuthill gear pump, and the mass flow rate was measured using an Endress+Hauser Promass A Coriolis flowmeter. Heating power was supplied to a cartridge heater mounted inside the inner cylindrical element of the test section and controlled by a programmable GW Instek ACR-2100 power supply.

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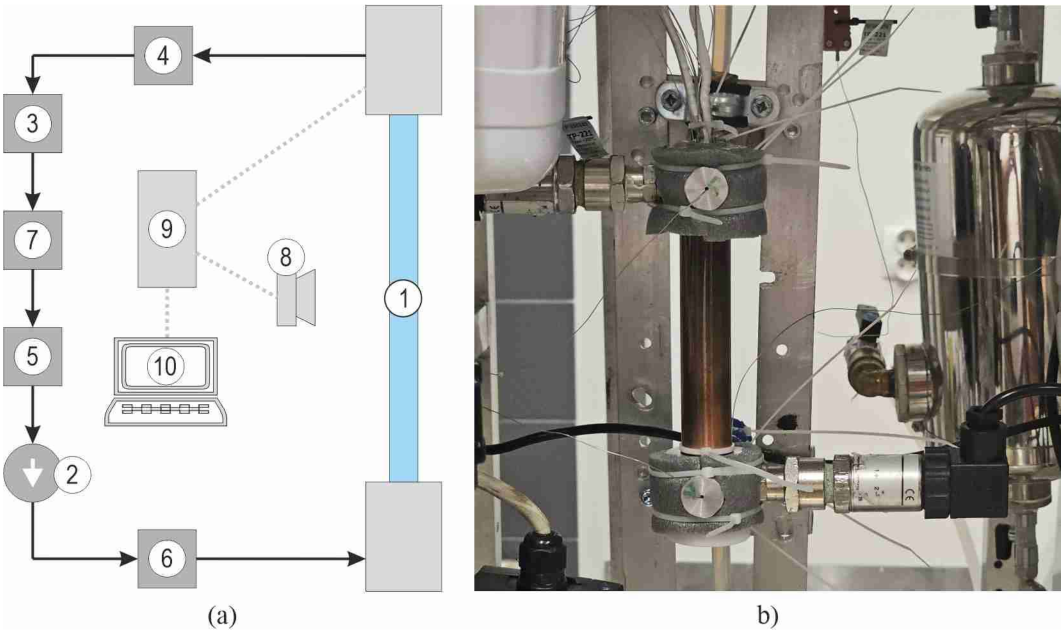

Figure 1. Experimental rig: (a) schematic diagram of the closed-loop circulation system; (b) photograph of the test-section assembly, including the annular-minichannel module and adjacent components of the experimental setup.

The annular minichannel was formed between the outer surface of an interchangeable heated metal tube and the inner surface of a coaxial borosilicate-glass tube. The test-module geometry is presented in Fig. 2a,b. Figure 2a shows the longitudinal schematic view with the axial positions of the thermocouples located in the heater–tube interspace. Figure 2b presents the cross-sectional schematic view of the annular-minichannel geometry, including the coaxial arrangement of the cartridge heater, heated metal tube and outer glass tube. Figure 2c shows a photograph of the test module assembly.

The cartridge-heater diameter was 15.0 mm. Ten thermocouples were installed within the heater-tube interspace, as shown in Fig. 2a. Their exact radial coordinates were not resolved individually; therefore, the measured temperatures were referred to a common effective radius of 7.5 mm in the simplified model. The outer diameter of the coated metal tube was 18.0 mm and the inner diameter of the glass tube was 20.4 mm, giving an annular gap width of 1.2 mm. The active heated length was 110 mm and the hydraulic diameter was 2.4 mm.

Fluid temperature and overpressure were measured at the inlet and outlet of the test section using T-type thermocouples and WIKA S-10 pressure transducers, respectively. Electrical parameters, fluid temperature, pressure and flow rate were recorded using the Graphtec GL840 data-acquisition system. The 1 s sampling interval enabled monitoring of the approach to steady conditions and averaging of the stabilised part of each measurement series. The main parameters of the experimental setup are listed in Table 1.

Table 1. Main parameters of the annular-minichannel experimental setup.

Description / value	Parameter
Distilled water	Working fluid
Annular gap between coaxial cylindrical surfaces	Minichannel geometry
15.0 mm	Cartridge-heater diameter
7.5 mm	Assumed thermocouple radius in the simplified mathematical model
18.0 mm	Outer diameter of heated metal tube
20.4 mm	Inner diameter of glass tube
1.2 mm	Minigap width
110 mm	Active minichannel length
0.0024 m	Hydraulic diameter
Borosilicate glass tube; optical access to the flow	Outer wall
Copper: 398 W/(m·K); stainless steel: 15.1 W/(m·K)	Substrate thermal conductivity
Tuthill gear pump	Circulation in the flow loop
Endress+Hauser Promass A Coriolis flowmeter	Flow measurement
T-type thermocouples at the minigap inlet and outlet, Czaki Thermo-Product	Fluid temperature measurement
WIKA S-10 pressure transducers at the minichannel test module inlet and outlet	Overpressure measurement
WIKA A-10 absolute pressure transmitter	Atmospheric pressure measurement
Graphtec GL840 data logger	Data acquisition
GW Instek ACR-2100	Power supply

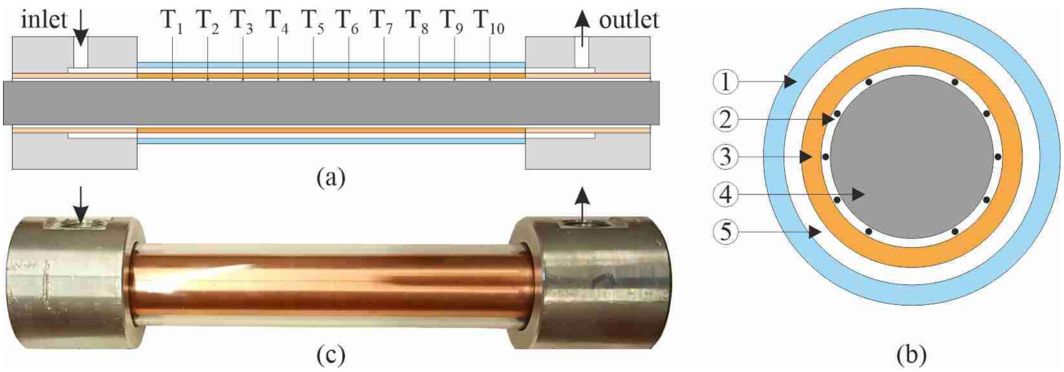

Figure 2. Annular-minichannel test module: (a) longitudinal schematic view showing the axial positions of the thermocouples located in the heater–tube interspace; (b) cross-sectional schematic view of the annular-minichannel geometry; (c) photograph of the test module assembly. 1 – glass tube; 2 – heater–tube interspace, locally accommodating the thermocouples; 3 – interchangeable heated metal tube with the tested surface modification; 4 – cartridge heater; 5 – annular minichannel.

3.2. Tested surfaces and coating preparation

The heated surface was prepared using interchangeable cylindrical tubes. Two substrate materials were considered: copper and stainless steel (hereafter referred to as steel). The copper matrix comprised smooth, ZnO and ZnO/PMHS surfaces. The steel matrix comprised smooth, ZnO, ZnO/PMHS and Al_2O_3 surfaces. The Al_2O_3 coating was thermally tested only on the steel substrate in the present experimental matrix, although a copper + Al_2O_3 specimen was included in the surface-characterisation set. For each substrate and nominal mass flow rate, the corresponding smooth tube was used as the differential reference. The tested thermal matrix is summarised in Table 2, and representative surface views are shown in Fig. 3.

Table 2. Tested substrate-surface matrix and coating codes used in the thermal experiments.

Role in the present thermal matrix	Coating code	Surface state	Substrate
Reference surface for the copper series	-	Smooth	Copper
ZnO-modified copper surface	A	ZnO	Copper
Low-surface-energy ZnO/PMHS-modified copper surface	C	ZnO/PMHS	Copper
Reference surface for the steel series	-	Smooth	Stainless steel
ZnO-modified steel surface	A	ZnO	Stainless steel
Low-surface-energy ZnO/PMHS-modified steel surface	C	ZnO/PMHS	Stainless steel
Ceramic-oxide coating tested thermally only on steel	B	Al_2O_3	Stainless steel

Figure 3. Representative views of the tested modified surfaces: ZnO and ZnO/PMHS coatings on copper and stainless steel substrates, and the Al_2O_3 coating on the steel substrate.

Preparation of ZnO coating (coating A). Polished samples were first immersed in a TEOS solution prepared by mixing 66 mL of isopropanol with 10 mL of tetraethyl orthosilicate (TEOS). Separately, 20 mL of isopropanol and 3 mL of acidified deionised water (pH 3-4, adjusted with HCl) were mixed and then slowly added to the TEOS solution under continuous stirring. The resulting solution was stirred for an additional 15-30 min. The samples were immersed in the prepared TEOS solution for approximately 30 s and then left to dry. Subsequently, the samples were immersed in a ZnO suspension prepared by dispersing 2 g of ZnO tetrapods in 100 mL of isopropanol followed by sonication for 15-30 min to ensure uniform particle distribution. Finally, the coated samples were heated at 100 °C for 1 h to promote crosslinking and stabilise the ZnO coating structure.

Preparation of Al_2O_3 coating (coating B). The preparation procedure was identical to that described for the ZnO coating. The only modification was the replacement of ZnO tetrapods with Al_2O_3 particles in the coating suspension. All subsequent processing steps, including TEOS pretreatment, particle deposition, drying and thermal treatment, were carried out under the same conditions as those used for the ZnO coating, resulting in the formation of an Al_2O_3 coating structure.

Preparation of ZnO/PMHS coating (coating C). The coating procedure was identical to that described for the ZnO coating until completion of the ZnO deposition and drying steps. The samples were then additionally immersed for 30 s in a 0.5 vol.% PMHS solution, followed by heat treatment at 100 °C for 1 h. The additional PMHS layer transformed the ZnO surface into a lower-surface-energy ZnO/PMHS coating.

The ZnO, ZnO/PMHS and Al_2O_3 coatings were not modelled as separate conductive layers because reliable coating thicknesses and effective thermal properties were not available for all specimens. Their influence was represented through the experimentally determined effective wall-fluid heat-transfer response. The separate contribution of coating resistance will be assessed when complete coating-thickness and thermal-property data become available.

3.3. Surface-characterisation methods

Surface roughness was characterised using a contact profilometer operating in stylus mode. Profile measurements were conducted over a 3000 μm scan length with a lateral resolution of approximately 1 μm. A stylus with a tip radius of 12.5 μm and a loading force of 3 mg was employed, while each scan required approximately 10 s. The average roughness parameter Ra was calculated from the recorded surface profiles. For each specimen, measurements were repeated at three independent positions, and the roughness values are reported as mean values with standard deviations where available.

Water contact-angle measurements were performed before and after the flow-boiling experiments. Static contact-angle values were used as surface descriptors. Because advancing and receding contact angles were not available for the present set, contact-angle hysteresis could not be evaluated.

The surface morphology of the coatings was characterised using a Hitachi S-3400N scanning electron microscope (SEM, Hitachi, Japan). SEM images were acquired at an accelerating voltage of 5 kV and a magnification of 5000×. Image analysis was performed with ImageJ software. SEM was used to document surface topography and morphology before and after the experiments and to support subsequent interpretation of differences in the thermal response.

3.4. Experimental procedure

Before each experimental series, the flow loop was filled with distilled water and deaerated. After the required mass flow rate had been set, the system was allowed to reach stable hydraulic and thermal conditions. Heater power was then increased stepwise to the maximum programmed value and subsequently decreased. Each operating point was held until the measured temperatures, pressure and flow rate were sufficiently stable for averaging.

Temperature, pressure, mass flow rate and heater-power data were recorded at 1 s intervals. The present data reduction used averaged values from the stabilised interval of each operating point. The increasing- and decreasing-power branches were retained separately for the pointwise comparison. Each operating point was treated as an individual steady experimental state; no interpolation between consecutive heat-flux settings was introduced.

No external heat-loss correction was applied in the present analysis because the outer glass-wall temperature was not measured during the water experiments. Consequently, the electrical heater power was used as the nominal heat input in the data-reduction model. In future experimental campaigns, the outer glass-wall temperature should be measured so that an external heat-loss term can be included in the energy balance without double-counting the heat transferred from the heater to the fluid.

4. Data-reduction model and heat transfer coefficient determination

A simplified local one-dimensional cylindrical-wall model was used to reconstruct the fluid-side wall temperature and the local heat transfer coefficient. The formulation follows the concentric-layer concept used previously for annular-minigap heat-transfer analysis [15], but the present reduction was restricted to radial conduction through the metal tube at each thermocouple position.

The electrical power P was assumed to pass radially through the metal tube of active length L, with a uniform axial distribution of heat input over L. The thermocouple temperature $T_{TC}(z)$ was referred to the assumed radius r_{TC} = 7.5 mm, while the wall-fluid interface was located at r_w = 9.0 mm. The thermal conductivity was taken as 398 W/(m·K) for copper and 15.1 W/(m·K) for stainless steel.

The fluid-side wall temperature was calculated from:

$$T_W(z) = T_{TC}(z) - \frac{P}{2\pi\, k_S\, L} ln\left(\frac{r_W}{r_{TC}}\right) \tag{1}$$

where k_s denotes the substrate thermal conductivity. Writing the correction in terms of total electrical power avoids ambiguity between heat fluxes referred to the inner and outer cylindrical surfaces.

The heat flux at the outer metal surface and the local heat transfer coefficient were calculated as:

$$q''_w = \frac{P}{(2\pi\, r_W\, L)} \tag{2}$$

$$\alpha(z) = \frac{q'\ w}{T_W(z) - T_{f,ref}(z)} \tag{3}$$

The local bulk-fluid temperature and local absolute pressure were interpolated linearly between the inlet and outlet measurements:

$$T_{f,bulk}(z) = T_{f,in} + (T_{f,out} - T_{f,in})\ \frac{z}{L} \tag{4}$$

$$p_{abs}(z) = p_{atm} + p_{in,g} + \left(p_{out,g} - p_{in,g}\right) \frac{z}{L} \tag{5}$$

The local saturation temperature was obtained from the water saturation relation, $T_{sat}(z) = T_{sat}[p_{abs}(z)]$ for water. In the present data set, $T_{f,bulk}$ was between 6.8 K and 125 K below T_{sat} at all analysed points. The complete series was therefore classified as bulk-subcooled, including later settings in which vapour bubbles were visually observed. Accordingly, $T_{f,ref} = T_{f,bulk}$ was used throughout the water analysis when calculating the local heat transfer coefficient.

The model does not distinguish a separate coating resistance. The calculated heat transfer coefficient should therefore be interpreted as an effective wall-fluid coefficient for the complete substrate-coating-fluid system under the stated assumptions.

5. Measurement uncertainty and comparison protocol

5.1. Measurement uncertainty

The present analysis is based on a comparative data-reduction procedure applied consistently to all smooth and modified surfaces. The principal measured quantities were electrical power, mass flow rate, inlet and outlet fluid temperatures, inlet and outlet pressures, and local thermocouple temperatures in the heated element. The accuracy of the measuring instruments is summarised in Table 3.

Table 3. Accuracy of the main measuring instruments used in the experimental setup.

Accuracy / uncertainty	Measuring instrument
±0.1% of measured value	Flow measurement (Coriolis mass flow meter)
±0.2°C (after calibration)	Temperature measurement (Type T thermocouple)
±0.5% of span	Overpressure measurement (pressure transducers)
±1% of span	Atmospheric pressure measurement (pressure transmitter)
±(0.5% of reading + 0.6 V)	Power supply – output voltage
±(0.5% of reading + 0.02 A)	Power supply – output current
±(2% of reading + 1 W)	Power supply – output active power (displayed)
±0.1% of reading	Data acquisition (Type T thermocouple)
±0.1% of full scale	Data acquisition (voltage input)
DAQ: ±0.1% of voltage-range full scale; shunt resistor: ±0.005% of reading	Data acquisition (voltage input with 250 Ω shunt resistor)

Because the study focuses on pointwise differences between modified surfaces and the corresponding smooth references, the same geometrical assumptions and data-reduction equations were used for each compared pair. This approach limits the influence of systematic modelling assumptions on the relative ranking of the tested surfaces.

A full propagated uncertainty analysis of the local heat transfer coefficient was not included at this stage. Therefore, the reported values are used primarily for comparative assessment of the tested surfaces under identical data-reduction assumptions. Future work will include a complete uncertainty analysis based on documented sensor accuracies, measured electrical power, wall-temperature measurements, and uncertainties of heat flux and heat transfer coefficient.

5.2. Differential comparison protocol

Each modified surface was compared with one designated complete smooth-reference series having the same substrate and nominal mass flow rate. For copper, earlier smooth runs were retained only for independent repeatability assessment. This procedure avoids averaging multiple reference runs with markedly different baselines into an artificial reference curve.

Pointwise pairs were formed at the same nominal heater power, the same occurrence of that power on the increasing or decreasing branch, and the same axial thermocouple position. For every pair, the local relative change in the heat transfer coefficient and the wall-temperature difference were calculated. The pointwise relative change was defined as the percentage difference between the modified-surface value and the corresponding smooth-reference value, using the smooth-surface value as the reference denominator. Both the mean and median of the local relative changes were retained; the median was used as a robustness indicator because local ratios can be sensitive to small temperature differences.

A modification was considered beneficial only when the heat transfer coefficient increased and the wall temperature decreased, or at least did not increase. The increasing- and decreasing-power branches were preserved in the matching procedure, and each steady-state operating point was treated as a separate experimental state.

6. Results

6.1. Surface descriptors: contact angle, roughness and SEM morphology

Figure 4 presents the water contact angles measured on samples with different coatings. Initially, the contact angles of the uncoated control samples were 105° for stainless steel and 90° for copper. For the stainless-steel sample coated with Al_2O_3, the contact angle remained unchanged at 70° before and after the experiment. A similar result was observed for the copper sample coated with Al_2O_3, where the contact angle changed only slightly from 73° before the experiment to 76° after the experiment.

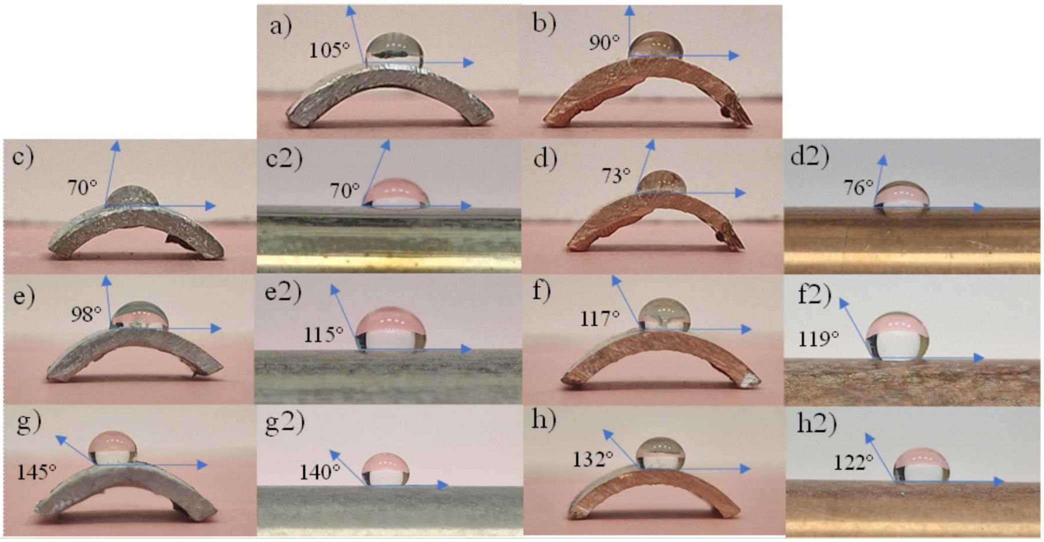

Figure 4. Water contact angles of the samples before the experiments: (a) Steel, (b) Copper, (c) Steel + Al_2O_3, (d) Copper + Al_2O_3, (e) Steel + ZnO, (f) Copper + ZnO, (g) Steel + ZnO/PMHS and (h) Copper + ZnO/PMHS; and after the experiments: (c2) Steel + Al_2O_3, (d2) Copper + Al_2O_3, (e2) Steel + ZnO, (f2) Copper + ZnO, (g2) Steel + ZnO/PMHS and (h2) Copper + ZnO/PMHS.

As shown in Fig. 4, the stainless-steel sample coated with ZnO exhibited a contact angle of 98° (Fig. 4e) before the experiment and 115° (Fig. 4e2) after the experiment. For the copper sample coated with ZnO, the corresponding values were 117° (Fig. 4f) and 119° (Fig. 4f2). The stainless-steel sample coated with ZnO/PMHS exhibited contact angles of 145° (Fig. 4g) before and 140° (Fig. 4g2) after the experiment, whereas the copper sample coated with ZnO/PMHS showed a decrease from 132° (Fig. 4h) to 122° (Fig. 4h2) . These measurements indicate that the final wetting state of the surfaces should be described using the measured contact-angle values rather than only the nominal coating type.

Figure 5 shows scanning electron microscope images of the samples coated with Al_2O_3, ZnO and ZnO/PMHS before and after the experiments, together with surface roughness values Ra. The uncoated stainless-steel and copper samples had relatively smooth surfaces, with Ra = 253.11 nm and Ra = 178.39 nm, respectively. After coating with Al_2O_3, an aluminium-oxide layer was visible on the surface. The stainless-steel Al_2O_3 sample exhibited Ra = 1143.63 nm before the experiment and Ra = 773.02 nm after the experiment, together with a non-uniform distribution of aluminium-oxide granules. The copper Al_2O_3 sample showed a smaller decrease in roughness, from Ra = 417.50 nm before the experiment to Ra = 369.64 nm after the experiment.

The ZnO-coated samples showed a marked increase in roughness after the experiments: from Ra = 536.9 nm to Ra = 1765.24 nm for stainless steel and from Ra = 1285.29 nm to Ra = 3742.68 nm for copper. In the SEM images, the ZnO tetrapod structures remained visible after exposure to water flow and temperature. A different roughness evolution was observed for ZnO/PMHS. The stainless-steel ZnO/PMHS sample decreased from Ra = 6755.10 nm before the experiment to Ra = 1121.21 nm after the experiment. The copper ZnO/PMHS sample decreased from Ra = 2254.39 nm to Ra = 1056.32 nm. The images show fine ZnO tetrapods covered by a PMHS layer and a post-test surface morphology that is visibly smoother than before the experiment.

The surface descriptors extracted from the available contact-angle and roughness results are summarised in Table 4.

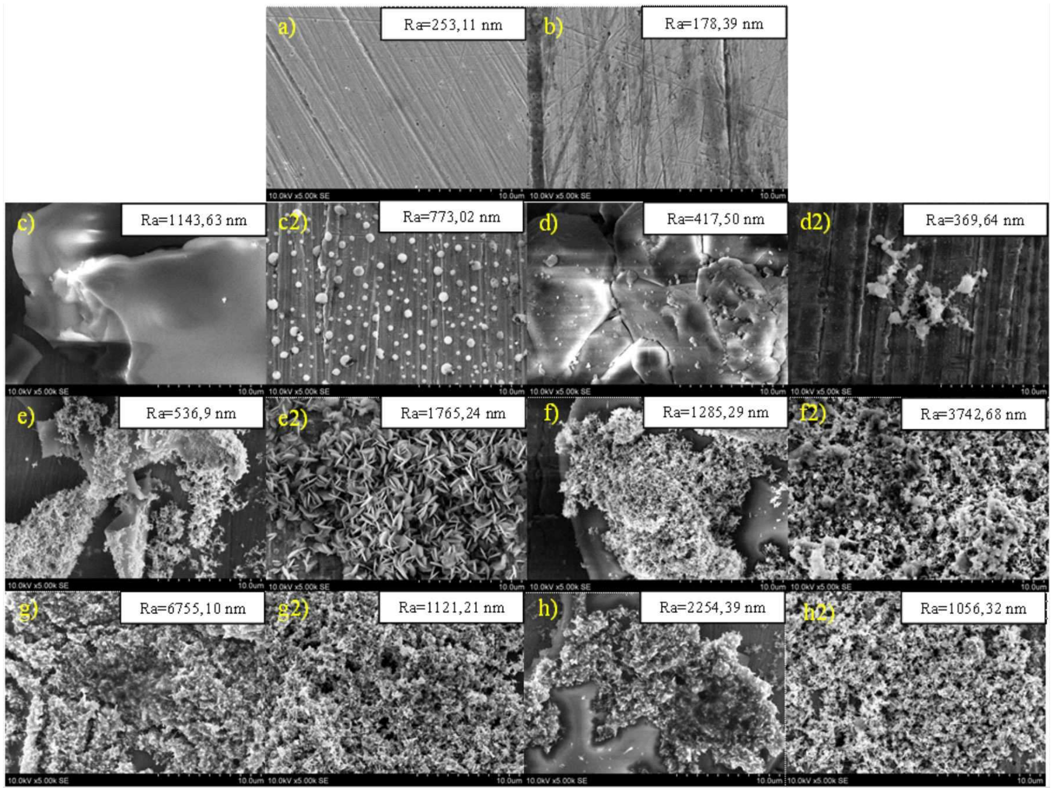

Figure 5. SEM images of the samples before the experiments with average roughness Ra: (a) Steel, (b) Copper, (c) Steel + Al_2O_3, (d) Copper + Al_2O_3, (e) Steel + ZnO, (f) Copper + ZnO, (g) Steel + ZnO/PMHS and (h) Copper + ZnO/PMHS; and after the experiments: (c2) Steel + Al_2O_3, (d2) Copper + Al_2O_3, (e2) Steel + ZnO, (f2) Copper + ZnO, (g2) Steel + ZnO/PMHS and (h2) Copper + ZnO/PMHS.

Table 4. Surface descriptors extracted from the available contact-angle and roughness results.

Comment	Ra before, nm	Ra after, nm	Contact angle after, °	Contact angle before, °	Tested	Specimen
Uncoated reference	253.11	—	—	105	Yes	Steel
Uncoated reference	178.39	—	—	90	Yes	Copper
Al_2O_3 coating tested thermally on steel	1143.63	773.02	70	70	Yes	Steel + Al_2O_3
Surface-characterisation specimen only	417.50	369.64	76	73	No	Copper + Al_2O_3
ZnO specimen; measured contact angle above 90°	536.9	1765.24	115	98	Yes	Steel + ZnO
ZnO specimen; measured contact angle above 90°	1285.29	3742.68	119	117	Yes	Copper + ZnO
Hydrophobic ZnO/PMHS specimen	6755.10	1121.21	140	145	Yes	Steel + ZnO/PMHS
Hydrophobic ZnO/PMHS specimen	2254.39	1056.32	122	132	Yes	Copper + ZnO/PMHS

6.2. General thermal response

The corrected bulk-subcooled analysis produced a clear separation between one favourable modification and the remaining surfaces. Figure 6 presents representative axial distributions of the local heat transfer coefficient for the steel surfaces at 290 W on the increasing-power branch, separately for nominal mass flow rates of 7 kg/h (Figure 6a) and 10 kg/h (Figure 6b). This figure shows how the local coefficient changes along the heated length and allows direct comparison between the smooth steel reference and the ZnO-, ZnO/PMHS- and Al_2O_3-modified steel surfaces under the same selected operating condition.

Figures 7 and 8 summarise the matched pointwise comparison for all tested modified surfaces. For each matched pair, the relative change in the local heat transfer coefficient was defined as the percentage difference between the modified-surface value and the corresponding smooth-reference value, normalised by the smooth-reference value. Figure 7 shows the mean and median relative changes in the local heat transfer coefficient with respect to the corresponding smooth reference of the same substrate and nominal mass flow rate. Figure 8 shows the corresponding mean wall-temperature change; negative values indicate that the modified surface operated at a lower wall temperature than the smooth reference. Taken together, Figs. 7 and 8 allow the heat-transfer coefficient response to be assessed together with the wall-temperature response, rather than treating coefficient enhancement alone as sufficient evidence of improvement.

In Figs. 6–8 and Table 5, steel + ZnO is the only modification that simultaneously increases the heat transfer coefficient and lowers the wall temperature at both nominal mass flow rates. Steel + Al_2O_3 and steel + ZnO/PMHS show the opposite combined response. The copper modifications produce no robust improvement relative to the designated smooth references. Absolute heat transfer coefficients varied substantially with axial position and experimental series; therefore, the principal interpretation is based on matched differential results rather than on the largest isolated local coefficient.

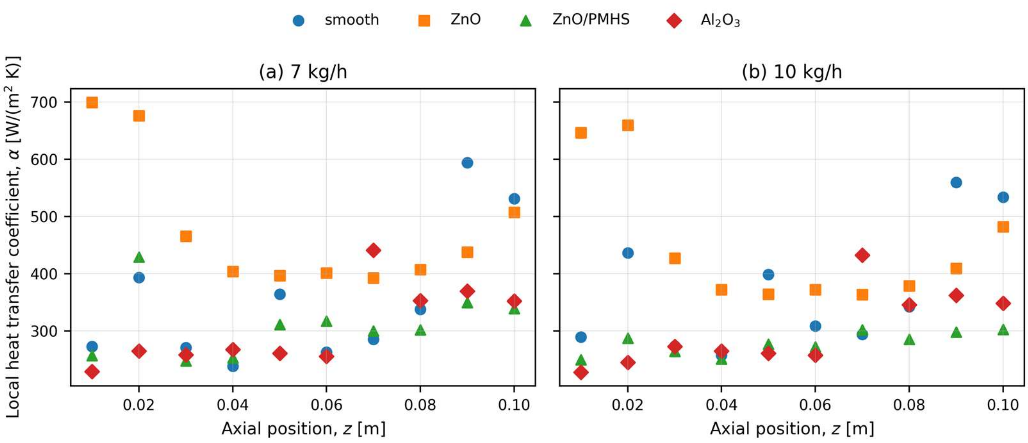

Figure 6. Local heat transfer coefficient along the steel tube for all tested steel surfaces at 290 W on the increasing-power branch: (a) 7 kg/h and (b) 10 kg/h.

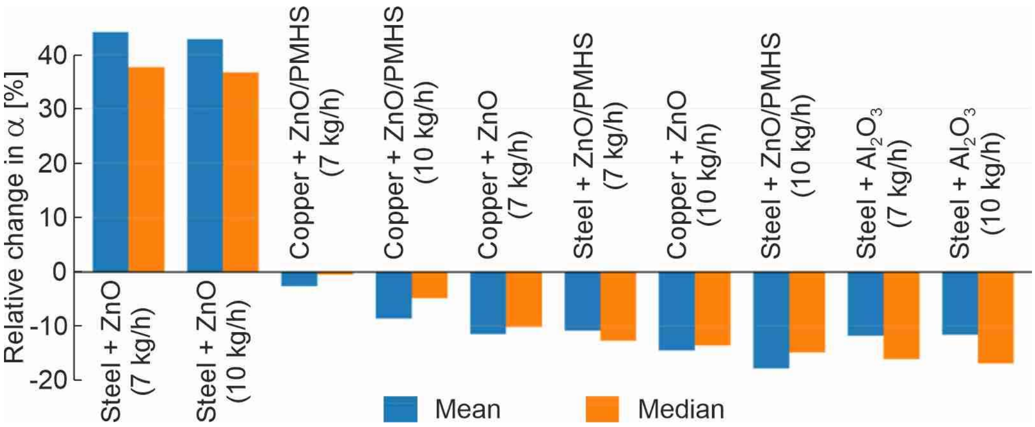

Figure 7. Mean and median pointwise relative changes in the local heat transfer coefficient with respect to the corresponding smooth reference.

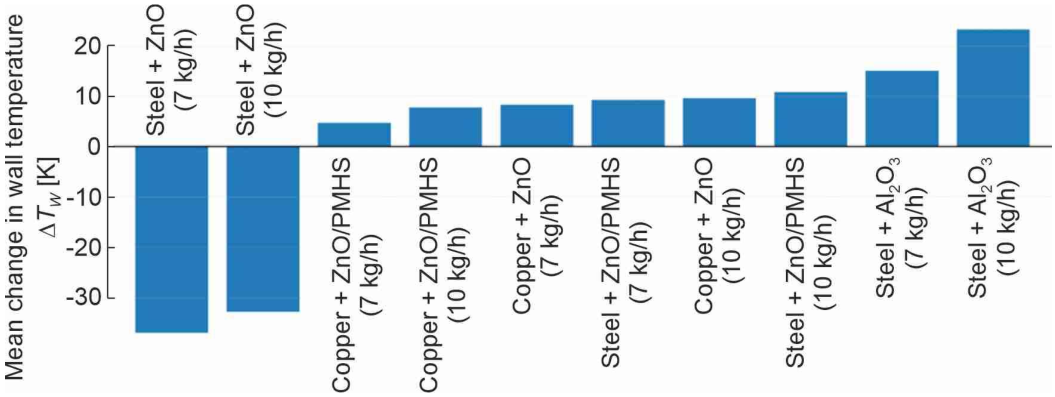

Figure 8. Mean wall-temperature change of each modified surface relative to the corresponding smooth reference. Negative values indicate a lower wall temperature.

Table 5. Bulk-subcooled differential performance of the modified surfaces relative to the corresponding smooth references.

Interpretation	Mean ΔT_w, K	Mean relative change (median), %	Average α, W/(m²·K)	Surface	Substrate and flow rate
Clear enhancement	-36.9 K	+44.2% (+37.8%)	438.8	ZnO	Steel, 7 kg/h
Clear enhancement	-32.8 K	+42.9% (+36.8%)	439.1	ZnO	Steel, 10 kg/h
Near-neutral α; no overall benefit	+4.7 K	-2.6% (-0.5%)	346.7	ZnO/PMHS	Copper, 7 kg/h
Unfavourable	+10.8 K	-11.4% (-10.1%)	310.0	ZnO	Copper, 7 kg/h
No enhancement	+8.3 K	-8.5% (-4.8%)	351.2	ZnO/PMHS	Copper, 10 kg/h
Unfavourable	+15.1 K	-14.4% (-13.5%)	313.9	ZnO	Copper, 10 kg/h
No enhancement	+7.8 K	-11.8% (-16.1%)	276.0	Al_2O_3	Steel, 7 kg/h
No enhancement	+9.6 K	-11.6% (-16.8%)	264.6	Al_2O_3	Steel, 10 kg/h
Unfavourable	+9.2 K	-10.7% (-12.7%)	273.9	ZnO/PMHS	Steel, 7 kg/h
Unfavourable	+23.3 K	-17.7% (-14.8%)	246.3	ZnO/PMHS	Steel, 10 kg/h

6.3. Relative heat-transfer performance of modified surfaces

The differential comparison used all matched points from the bulk-subcooled formulation. For each modified surface, the corresponding smooth run was matched by nominal heater-power setting, power branch, and axial position. Table 5 reports the average local heat transfer coefficient of the modified surface, the mean and median pointwise relative changes, and the mean wall-temperature difference. The same ranking is visualised in Figs. 7 and 8.

The mean pointwise percentage change and the ratio of global means are not identical quantities. The former preserves the contribution of every matched local point, whereas the latter can be dominated by regions with larger absolute coefficients. The present ranking therefore reports the mean pointwise change together with its median.

When Table 5 is analysed together with Figs. 7 and 8, steel + ZnO is the only surface that simultaneously increases the local heat transfer coefficient and lowers the wall temperature at both analysed flow rates. The mean pointwise enhancement exceeds 42%, while the corresponding medians are approximately 37%. The Al_2O_3 coating does not reproduce this effect: the coefficient decreases by about 12% on average and the wall temperature increases by 8-10 K. The ZnO/PMHS steel coating also performs unfavourably, especially at 10 kg/h.

6.4. Steel surfaces

The steel substrate showed the clearest coating-dependent response. In Fig. 6, the ZnO-modified steel surface exhibits higher local coefficients over most of the heated length at both nominal mass flow rates. The aggregate comparison in Table 5 and Figs. 7 and 8 confirms this trend: the average local heat transfer coefficient was 438.8 W/(m²K) at 7 kg/h and 439.1 W/(m²K) at 10 kg/h, while the corresponding mean pointwise gains relative to smooth steel were 44.2% and 42.9%, with medians of 37.8% and 36.8%. Mean wall temperature decreased by 36.9 K and 32.8 K, respectively.

Neither ZnO/PMHS nor Al_2O_3 improved the steel reference. ZnO/PMHS reduced the mean local coefficient by 10.7% at 7 kg/h and 17.7% at 10 kg/h, while increasing wall temperature by 9.2 K and 23.3 K. Al_2O_3 reduced the mean local coefficient by approximately 11.6-11.8% and increased wall temperature by approximately 7.8-9.6 K. Thus, the favourable steel response was specific to ZnO rather than to oxide modification in general.

6.5. Copper surfaces and reference repeatability

The copper modifications did not provide a robust benefit relative to the designated smooth references. Copper + ZnO reduced the mean local coefficient by 11.4% at 7 kg/h and 14.4% at 10 kg/h, while increasing mean wall temperature by 10.8 K and 15.1 K. Copper + ZnO/PMHS was closer to neutral at 7 kg/h, with a mean pointwise change of -2.6% and a median of -0.5%, but wall temperature still increased by 4.7 K. At 10 kg/h, copper + ZnO/PMHS reduced the mean local coefficient by 8.5% and increased wall temperature by 8.3 K.

The repeatability of the independent smooth-copper reference runs is illustrated in Fig. 9 by local heat transfer coefficient profiles as a function of axial position at a heater power of 290 W and at two nominal mass flow rates. The run-to-run discrepancies shown in Fig. 9 are comparable to or greater than the apparent effects of the copper coatings. Therefore, none of the copper coatings can currently be classified as heat-transfer enhancing. These conclusions apply only to the present distilled-water experiments and the selected smooth-reference runs. The response may differ in the planned experiments using refrigerants as working fluids, because their viscosity, surface tension, latent heat, saturation pressure and wetting behaviour differ substantially from those of water.

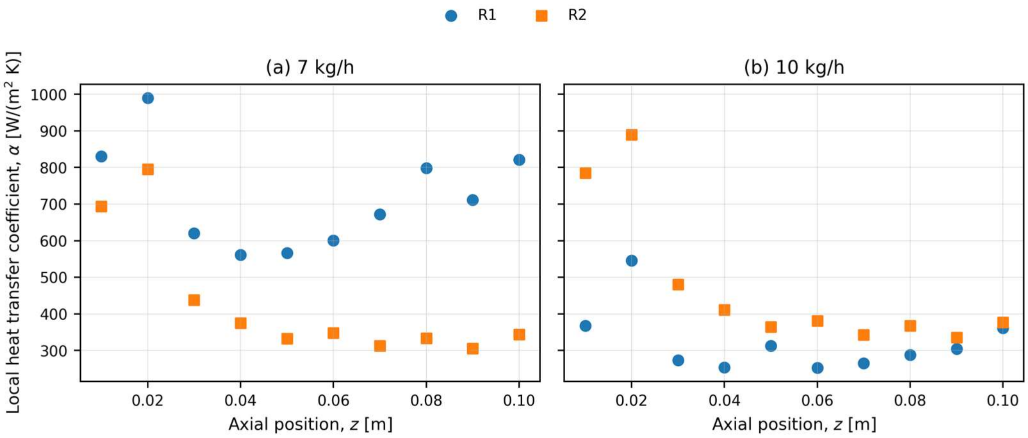

Figure 9. Repeatability of independent smooth copper reference runs illustrated by local heat transfer coefficient profiles at 290 W on the increasing-power branch: (a) 7 kg/h and (b) 10 kg/h. R1 and R2 denote separate experimental runs.

7. Discussion

The present analysis provides a comparative assessment of the tested coating-substrate combinations under identical data-reduction assumptions. Among the investigated surfaces, ZnO on stainless steel produced the most consistent favourable response, combining an increase in the local heat transfer coefficient with a decrease in wall temperature. This result indicates that the thermal effect of the surface modification depends not only on the coating material itself, but also on its interaction with the substrate and the working fluid.

For steel + ZnO, the improvement was obtained relative to the smooth steel reference, using the same substrate thermal conductivity and the same cylindrical temperature correction. The observed differential gain can therefore be attributed to the modified surface condition rather than to the bulk thermal conductivity of the substrate alone. The available surface descriptors suggest that the response may be related to the morphology and stability of the ZnO layer, the distribution of active nucleation sites, liquid replenishment near the wall and bubble departure behaviour. These effects are consistent with the simultaneous increase in the local heat transfer coefficient and reduction in wall temperature.

The contact-angle results also show that the final wetting state should be described using measured values rather than only nominal coating labels. The ZnO-coated specimens exhibited static contact angles above 90° after preparation and a further increase after the experiments. Therefore, the favourable steel + ZnO response should not be interpreted simply as the effect of a nominally hydrophilic coating. Instead, it should be treated as the combined result of the measured wetting state, surface morphology, coating-substrate interaction and local boiling conditions.

The results obtained for steel + Al_2O_3 and steel + ZnO/PMHS confirm that surface modification does not automatically improve boiling heat transfer. The Al_2O_3 coating did not reproduce the beneficial response observed for ZnO on steel, although it was also a ceramic-oxide surface. Similarly, the ZnO/PMHS coating reduced the local heat transfer coefficient and increased wall temperature, especially at the higher nominal mass flow rate. This behaviour may result from changes in liquid-solid contact, vapour retention within the surface structure or additional thermal resistance associated with the modified surface layer.

The copper results should be interpreted with particular care because the repeated smooth copper reference runs showed noticeable baseline variability. Under the present comparison protocol, neither ZnO nor ZnO/PMHS on copper provided a robust improvement relative to the designated smooth references. Additional repetitions would be required before drawing stronger conclusions about the copper-substrate variants.

All analysed water data remained bulk-subcooled, even when vapour bubbles were observed at higher heat inputs. The reported results therefore describe subcooled flow boiling in the present annular minichannel rather than saturated bulk flow. Future experiments with refrigerants, dielectric liquids and cooling fluids will be used to verify whether the favourable ranking of steel + ZnO persists when fluid properties, saturation pressure, wetting behaviour and bubble dynamics differ from those of distilled water.

8. Future work

The present water series provides a self-contained baseline for the next experimental campaign. The principal methodological priorities are improved characterisation of the heat input and external heat losses, direct documentation of the thermocouple radius, and a defined repeatability protocol for each smooth and modified surface.

The next experimental campaign should use refrigerants, dielectric liquids and cooling fluids together with a better-defined measurement chain. The outer glass-wall temperature should be recorded, the radial thermocouple location should be documented, and measured voltage-current power should be retained together with programmed power. Surface characterisation should include static, advancing and receding contact angles, contact-angle hysteresis, roughness, coating thickness, morphology and coating stability before and after boiling tests.

The repeatability protocol will include additional runs for both smooth and modified surfaces. Uncertainty propagation will be applied to heat flux, corrected wall temperature, local bulk temperature, local saturation temperature and local heat transfer coefficient.

Future experimental investigations using annular-minichannel test modules should employ a geometry-specific data-reduction procedure that accounts for heat losses through the glass wall when the required wall-temperature measurements are available. Once reproducible refrigerant data series have been obtained, the annular- and rectangular-minichannel databases will be linked using common dimensionless parameters. The combined database will then be used to develop and validate machine-learning models for predicting heat-transfer performance and identifying coating–substrate–fluid combinations that provide a reproducible thermal advantage, together with the operating ranges over which this advantage persists.

9. Conclusions

An experimental study of subcooled flow boiling of distilled water in an annular minichannel was performed for smooth, ZnO-, ZnO/PMHS- and Al_2O_3-modified heated tubes. Copper and steel substrates were analysed using a corrected cylindrical data-reduction model and pointwise differential comparison with designated smooth references.

All analysed local bulk temperatures remained below the corresponding local saturation temperatures by at least 6.8 K. The present water series was therefore treated as bulk-subcooled, and $T_{f,bulk}$ was used as the reference temperature in the local heat transfer coefficient. Vapour bubbles observed at later settings represented subcooled boiling rather than a saturated bulk-flow state.

The ZnO-modified steel surface produced the strongest and only robust positive response. The mean pointwise relative increase in the heat transfer coefficient was 44.2% at 7 kg/h and 42.9% at 10 kg/h, with medians of 37.8% and 36.8%. Mean wall temperature decreased by 36.9 K and 32.8 K, respectively. The local profiles and differential rankings confirm that this result is not limited to isolated points.

The Al_2O_3-modified steel surface did not enhance heat transfer. Its mean local coefficient decreased by approximately 12%, while wall temperature increased by approximately 8-10 K. Steel + ZnO/PMHS also performed unfavourably, particularly at 10 kg/h. Thus, the beneficial response was specific to ZnO on steel and cannot be generalised to all oxide or low-surface-energy coatings.

The copper modifications produced no robust improvement. Their interpretation is additionally limited by large discrepancies between repeated smooth copper runs. One designated smooth run was therefore used for each principal comparison, while the earlier smooth runs were retained only for repeatability assessment. The present distilled-water experiments establish a reference data set for future investigations with refrigerants, dielectric liquids and cooling fluids.

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Conflicts of Interest: The authors declare no conflicts of interest.

Nomenclature

Unit	Description	Symbol
W/(m·K)	Thermal conductivity of the substrate material	k_s
m	Active heated length of the test section	L
W	Electrical heater power used in the data-reduction model	P
Pa	Local absolute pressure in the minichannel	p_{abs}
Pa	Atmospheric pressure	p_{atm}
Pa	Gauge pressure measured at the test-section inlet	$p_{in,g}$
Pa	Gauge pressure measured at the test-section outlet	$p_{out,g}$
W/m²	Heat flux referred to the outer surface of the heated metal tube	q''_w
nm or µm	Arithmetic mean surface roughness	R_a
m	Assumed radial position of the thermocouple temperature in the simplified model	r_{TC}
m	Outer radius of the heated metal tube; wall-fluid interface radius	r_w
°C or K	Local bulk-fluid temperature	$T_{f,bulk}$
°C or K	Fluid temperature measured at the test-section inlet	$T_{f,in}$
°C or K	Fluid temperature measured at the test-section outlet	$T_{f,out}$
°C or K	Reference fluid temperature used in the heat-transfer-coefficient definition	$T_{f,ref}$
°C or K	Local saturation temperature calculated from local absolute pressure	T_{sat}
°C or K	Thermocouple temperature measured in the heated element	T_{TC}

Unit	Description	Symbol
°C or K	Calculated local wall temperature at the heated-tube–fluid interface	T_w
m	Axial coordinate measured from the test-section inlet	z
Greek		
W/(m²·K)	Local heat transfer coefficient at the wall-fluid interface	α
%	Pointwise relative change in local heat transfer coefficient with respect to the corresponding smooth reference	$\Delta\alpha/\alpha_{ref}$
K	Pointwise or mean wall-temperature change relative to the corresponding smooth reference	ΔT_w
Abbreviations and coating names		
Aluminium oxide		Al_2O_3
Polymethylhydrosiloxane		PMHS
Scanning electron microscopy		SEM
Tetraethyl orthosilicate		TEOS
Zinc oxide		ZnO
ZnO coating modified with PMHS		ZnO/PMHS

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